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Food Rescue Connect: Smart Food Donation and Redistribution Management System

Industry Context

Agile sprint planning & user story workshop

A CASE STUDY REPORT

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1 INTRODUCTION:

This document presents an Agile Sprint Planning and User Story Workshop conducted for the project "Food Rescue Connect: Smart Food Donation and Redistribution Management System." The purpose of this exercise is to apply agile project management techniques - including vision definition, epic and user story creation, acceptance criteria writing, backlog prioritization, sprint planning, and story point estimation - to a real-world software problem statement.

Food Rescue Connect is envisioned as an intelligent digital platform that connects food donors such as restaurants, hotels, supermarkets, and event organizers with NGOs, volunteers, and beneficiaries who require surplus food. The platform leverages Artificial Intelligence, Cloud Computing, GPS, and IoT to prioritize donations, optimize logistics, and ensure transparent, timely food redistribution.

The following sections apply the Scrum framework - starting from the product vision, moving through backlog construction, and culminating in a fully planned first sprint with story-point estimates and a clear sprint goal. Each artifact has been developed by directly referencing the functional requirements, modules, and workflow identified in the case study, ensuring traceability between the business problem and the agile plan.

1.1 Agile Approach Adopted

- Framework: Scrum, with two-week sprint cycles.
- Estimation technique: Planning Poker using the Fibonacci sequence (1, 2, 3, 5, 8, 13, 21).
- Prioritization technique: MoSCoW method (Must have, Should have, Could have, Won't have) supported by business-value scoring.
- Ceremonies: Sprint Planning, Daily Stand-up, Sprint Review, and Sprint Retrospective.

2. Project Vision and Stakeholder Identification

2.1 Project Vision

To build a centralized, AI-driven digital ecosystem that eliminates communication gaps between surplus food donors and hungry communities - reducing food waste, minimizing food spoilage, and ensuring that safe, nutritious food reaches vulnerable populations quickly, transparently, and equitably, while supporting environmental sustainability and the United Nations Sustainable Development Goal of Zero Hunger.

2.2 Project Objectives

- Provide a single centralized platform for donors, NGOs, volunteers, and beneficiaries to coordinate food donation in real time.
- Use AI to prioritize donations by expiry time, quantity, demand, and distance so that food is collected before it spoils.
- Use GPS to compute optimal collection and delivery routes, minimizing transportation time and cost.
- Offer real-time notifications and live tracking to all stakeholders for full transparency and accountability.
- Generate analytical reports on donation volume, meals served, volunteer performance, and food-waste reduction.
- Reduce landfill waste and greenhouse-gas emissions associated with discarded food.

2.3 Stakeholder Identification

The success of Food Rescue Connect depends on the coordinated participation of the following stakeholder groups, each with distinct roles and expectations from the system:

Stakeholder	Role	Key Interest / Expectation
Food Donors	Restaurants, hotels, supermarkets, caterers, and event organizers who register surplus food	Quick, hassle-free listing and confirmation that donated food reaches beneficiaries
NGOs / Charitable Organizations	Receive, verify, and distribute donated food to beneficiaries	Timely, relevant donation alerts matched to their capacity and beneficiary demand
Volunteers	Collect food from donors and deliver it to NGOs / beneficiaries	Clear pickup/delivery tasks, optimized routes, and manageable workload
Beneficiaries	Individuals and communities receiving redistributed food	Reliable and safe access to nutritious food

Stakeholder	Role	Key Interest / Expectation
System Administrators	Manage platform operations, user verification, and system configuration	Stable, secure, and well-monitored platform
Government / Regulatory Bodies	Provide food-safety guidelines and support large-scale coordination	Compliance with food safety and public health regulations
Logistics / Delivery Partners	Provide transportation support for large donation volumes	Efficient route information and scheduling
Project Sponsors / Institution	Fund and oversee the development of the platform	On-time, on-budget delivery of a scalable, impactful solution

3. Epics and User Stories

The product requirements identified in the case study have been decomposed into six Epics. Each Epic is further broken down into detailed User Stories following the standard Agile format: "As a <user>, I want <feature>, so that <benefit>."

Epic 1: Donor Food Registration & Management

Enables food donors to register, describe, and manage surplus food listings.

US-01 As a food donor, I want to register surplus food with type, quantity, expiry time, and pickup location, so that NGOs and volunteers can be notified immediately.

US-02 As a food donor, I want to edit or cancel a donation listing before it is picked up, so that I can correct mistakes or withdraw food that is no longer available.

US-03 As a food donor, I want to receive confirmation once my donation has been collected, so that I know my contribution was successfully redirected.

US-04 As a frequent donor (restaurant/hotel), I want to save recurring donation templates, so that I can list regular surplus food faster.

Epic 2: AI-Based Matching & Allocation

Uses Artificial Intelligence to prioritize donations and match them to the most suitable beneficiary organization.

US-05 As a system, I want to automatically prioritize donations by expiry time, quantity, and distance, so that the most urgent food is collected first and spoilage is minimized.

US-06 As a NGO coordinator, I want to receive AI-recommended donation matches based on my organization's current demand, so that I can accept the most relevant donations without manual searching.

US-07 As a system administrator, I want to configure the weighting of AI prioritization factors, so that the allocation logic can be tuned as real-world conditions change.

Epic 3: Volunteer & Logistics (GPS) Management

Assigns volunteers to donations and computes optimal collection/delivery routes.

US-08 As a volunteer, I want to view nearby available pickup requests on a map, so that I can choose tasks that fit my location and schedule.

US-09 As a volunteer, I want to receive the GPS-optimized route for pickup and delivery, so that I can complete the task in the shortest possible time.

US-10 As a system, I want to automatically assign the nearest available volunteer to an urgent donation, so that collection delays are minimized.

Epic 4: Real-Time Tracking & Notifications

Keeps all stakeholders informed with live status updates throughout the donation lifecycle.

US-11 As a NGO coordinator, I want to track the live status of an assigned pickup, so that I can plan receiving staff and storage in advance.

US-12 As a food donor, I want to receive push notifications when my donation is accepted, collected, and delivered, so that I have full visibility into what happened to my contribution.

US-13 As a beneficiary, I want to receive an alert when food is about to be delivered to my community centre, so that I can be present to receive it.

Epic 5: Reporting & Analytics

Provides administrators and NGOs with insight into donation trends, volunteer performance, and impact.

US-14 As an administrator, I want to view a dashboard of total food donated, meals served, and waste reduction, so that I can evaluate the platform's overall impact.

US-15 As a NGO manager, I want to generate monthly reports of donations received and distributed, so that I can share impact summaries with my own stakeholders.

US-16 As an administrator, I want to view volunteer performance metrics, so that I can recognize top contributors and identify areas needing more volunteers.

Epic 6: User Account & Security Management

Provides secure registration, authentication, and profile management for all user roles.

US-17 As a new user, I want to register and verify my account (donor, NGO, volunteer, or beneficiary), so that I can securely access role-appropriate features.

US-18 As a registered user, I want to reset my password securely, so that I can regain access if I forget my credentials.

US-19 As a administrator, I want to verify and approve new NGO accounts before they can accept donations, so that only legitimate organizations participate on the platform.

4. Acceptance Criteria

Acceptance criteria have been written for the highest-priority user stories selected for Sprint 1, using the Given-When-Then format to ensure each story is specific, measurable, and testable.

US-01 - Donor registers surplus food

- Given a logged-in food donor is on the "New Donation" screen
- When the donor submits food type, quantity, expiry time, and pickup location
- Then the system saves the donation, assigns it a unique ID, and notifies nearby NGOs and volunteers within 60 seconds

US-05 - AI prioritizes donations

- Given multiple active donations exist with different expiry times, quantities, and distances
- When the AI allocation engine runs its prioritization routine
- Then donations are ranked by urgency (soonest expiry first) and the ranked list is visible to NGOs and administrators

US-06 - NGO receives AI-recommended matches

- Given an NGO's profile specifies its current food demand and capacity
- When a new donation matching that demand is registered
- Then the NGO receives a recommendation notification with a match-relevance score above the configured threshold

US-08 - Volunteer views nearby pickups

- Given a volunteer is logged in with location services enabled
- When the volunteer opens the "Available Pickups" map view
- Then all unassigned donation requests within the volunteer's configured radius are displayed with distance and expiry time

US-09 - Volunteer receives GPS-optimized route

- Given a volunteer has accepted a pickup and delivery task

- When the volunteer taps "Start Route"
- Then the system displays the shortest path from the volunteer's location to the donor, then to the NGO/beneficiary, with estimated time of arrival

US-11 - NGO tracks live pickup status

- Given a volunteer has been assigned to collect an accepted donation
- When the volunteer updates status (en route, collected, delivering, delivered)
- Then the NGO coordinator's dashboard reflects the updated status within 30 seconds

US-12 - Donor receives status notifications

- Given a donor's listed donation changes status
- When the donation is accepted, collected, or delivered
- Then the donor receives a push/email notification corresponding to each status change

US-14 - Administrator views impact dashboard

- Given donation, collection, and delivery records exist in the system
- When an administrator opens the analytics dashboard
- Then total food donated (kg), meals served, active donors, and estimated CO2/landfill waste avoided are displayed, filterable by date range

US-17 - New user registers an account

- Given a new user is on the registration page
- When the user selects a role (donor, NGO, volunteer, beneficiary) and submits valid details
- Then the account is created in a "pending verification" state and a confirmation email/OTP is sent

5. Product Backlog Prioritization

The product backlog below lists all identified user stories, prioritized using the MoSCoW technique and a business-value score (1-10). Higher business value and Must-have priority stories were selected first for Sprint 1.

ID	User Story (short)	Priority	Business Value	Story Pts	Sprint
US-01	Donor registers surplus food	Must	10	5	Sprint 1
US-05	AI prioritizes donations	Must	10	8	Sprint 1
US-08	Volunteer views nearby pickups	Must	9	5	Sprint 1
US-09	GPS-optimized route for volunteer	Must	9	8	Sprint 1
US-11	NGO tracks live pickup status	Must	8	5	Sprint 1
US-17	New user registration & verification	Must	9	5	Sprint 1
US-12	Donor status notifications	Should	7	3	Sprint 1
US-06	NGO receives AI-recommended matches	Should	8	8	Sprint 2
US-02	Donor edits/cancels listing	Should	6	3	Sprint 2
US-10	Auto-assign nearest volunteer	Should	7	8	Sprint 2
US-13	Beneficiary delivery alert	Should	6	3	Sprint 2
US-18	Password reset	Should	5	2	Sprint 2
US-19	Admin verifies NGO accounts	Must	8	3	Sprint 1
US-14	Admin impact dashboard	Could	7	8	Sprint 3
US-15	NGO monthly reports	Could	5	5	Sprint 3
US-16	Volunteer performance metrics	Could	4	5	Sprint 3
US-03	Donor collection confirmation	Could	4	2	Sprint 3
US-04	Recurring donation templates	Won't (this release)	3	3	Backlog
US-07	Admin tunes AI weighting	Won't (this release)	3	5	Backlog

Prioritization rationale: stories enabling the core donation-to-delivery flow (registration, AI prioritization, volunteer discovery, GPS routing, live tracking, and account verification) were classified as Must-have because the platform cannot function without them.

6. Sprint Backlog Planning

Sprint 1 (Duration: 2 weeks) focuses on delivering the core donation registration, AI prioritization, volunteer discovery, GPS routing, live tracking, and account-verification flow - the minimum end-to-end path from a donor listing food to a volunteer completing delivery. Each selected user story has been broken down into actionable development tasks.

User Story	Development Tasks
US-01: Donor registers surplus food	Design donation form UI; build donation database schema; implement submit API; trigger notification service; write unit tests
US-05: AI prioritizes donations	Define prioritization scoring model (expiry, quantity, distance); implement scoring service; integrate with donation feed; validate with sample data
US-08: Volunteer views nearby pickups	Build map view component; implement geolocation-based query API; display distance/expiry on markers; test on mobile viewport
US-09: GPS-optimized route	Integrate mapping/routing API; compute shortest multi-stop route; render turn-by-turn view; display ETA
US-11: NGO tracks live pickup status	Build status state-machine (accepted/en route/collected/delivered); implement real-time update channel; build NGO dashboard status widget
US-17: User registration & verification	Build registration form with role selection; implement OTP/email verification service; set account status flow; write security tests for input validation
US-12: Donor status notifications	Integrate push/email notification service; map status changes to notification templates; test delivery timing
US-19: Admin verifies NGO accounts	Build admin review queue UI; implement approve/reject API; log verification decisions; notify NGO of outcome

6.1 Sprint 1 Capacity Summary

- Team size: 5 (2 backend, 2 frontend, 1 QA/DevOps)
- Sprint duration: 10 working days (2-week sprint)
- Total story points committed: 34 points
- Velocity assumption (first sprint, conservative baseline): 30-35 points

7. Story Point Estimation

Story points were estimated using Planning Poker with the Fibonacci sequence (1, 2, 3, 5, 8, 13, 21), where each number represents relative effort, complexity, and uncertainty rather than absolute hours. The development team independently proposed estimates, discussed differences, and converged on a consensus value for each story.

ID	User Story (short)	Points	Justification
US-01	Donor registers surplus food	5	Standard CRUD form with validation and a notification trigger; moderate complexity
US-05	AI prioritizes donations	8	Requires designing and validating a multi-factor scoring algorithm; higher uncertainty
US-08	Volunteer views nearby pickups	5	Map integration and geo-query logic; moderate complexity, some third-party API risk
US-09	GPS-optimized route	8	Multi-stop route optimization and third-party routing API integration; higher complexity
US-11	NGO tracks live pickup status	5	Requires real-time update mechanism (websocket/polling) and state-machine logic
US-17	User registration & verification	5	Multi-role registration with OTP/email verification and security validation
US-12	Donor status notifications	3	Reuses existing notification service; low complexity, mainly template/mapping work
US-19	Admin verifies NGO accounts	3	Simple approval workflow with moderate UI and logging needs

Fibonacci spacing was deliberately chosen: as effort and uncertainty grow, precise estimation becomes harder, so the gaps between higher numbers (5 -> 8 -> 13) are wider than between lower numbers (1 -> 2 -> 3). This discourages false precision on larger, less-understood stories such as US-05 (AI scoring) and US-09 (route optimization), which carry the highest technical uncertainty in Sprint 1.

8. Sprint Goal and Expected Deliverables

8.1 Sprint Goal

"By the end of Sprint 1, a donor shall be able to register a surplus food donation, have it automatically prioritized by the AI engine, matched to a nearby verified NGO, and collected and delivered by a volunteer using an optimized GPS route - with the NGO able to track the delivery status live from acceptance to completion."

8.2 Expected Sprint Deliverables

- Functional donor registration module with validation and database persistence.
- Working AI prioritization engine ranking active donations by urgency.
- Volunteer map view showing nearby, prioritized pickup requests.
- GPS-based route computation from donor to NGO/beneficiary with ETA.
- Real-time status tracking visible on the NGO dashboard.
- Secure multi-role user registration and account verification workflow.
- Donor notifications for key status changes (accepted, collected, delivered).
- Admin workflow for verifying and approving new NGO accounts.

8.3 Definition of Done

- Code reviewed, merged, and deployed to the staging environment.
- Unit and integration tests written and passing for each story.
- Acceptance criteria for each story verified by QA.
- No critical or high-severity defects open against the story.
- Feature demonstrated during the Sprint Review.

9. Reflection

Conducting this Agile Sprint Planning and User Story Workshop for Food Rescue Connect strengthened my understanding of how abstract project requirements are translated into structured, actionable agile artifacts. Breaking the case study's problem statement down into epics and user stories required me to think from the perspective of each stakeholder - donor, NGO coordinator, volunteer, beneficiary, and administrator - which improved my ability to capture requirements that are specific, testable, and value-driven.

Writing acceptance criteria in the Given-When-Then format taught me to think about edge cases and measurable outcomes rather than vague feature descriptions. Prioritizing the backlog using MoSCoW and business value forced me to make deliberate trade-offs about what constitutes a minimum viable end-to-end flow versus features that add value later. Estimating story points using the Fibonacci sequence also helped me appreciate how uncertainty, not just effort, should influence estimation, particularly for AI- and GPS-dependent stories.

Overall, this exercise improved my practical skills in agile project management, stakeholder analysis, and sprint planning, and gave me confidence in applying these techniques to real software engineering projects in the future.

10. Conclusion

This Agile Sprint Planning and User Story Workshop has translated the Food Rescue Connect case study into a complete, actionable agile plan. Starting from a clear project vision and stakeholder map, the requirements were decomposed into six epics and nineteen user stories, each supported by measurable acceptance criteria. The backlog was prioritized using MoSCoW and business value, and a realistic first sprint was planned - covering donation registration, AI-based prioritization, volunteer and GPS logistics, live tracking, notifications, and account verification - with story points estimated through Planning Poker.

The resulting sprint plan provides a clear, testable, and incrementally deliverable path toward building Food Rescue Connect, demonstrating how agile methodologies can be applied to manage a socially impactful software project effectively from vision through to sprint execution.

11. References

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